Equilibrium and missing forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a missing force when acceleration is zero.

SUCCESS CRITERIA - I can...

1. Recognize rest and constant velocity as possible equilibrium.
2. Find missing forces from a zero net-force condition.
3. Explain why turning is not equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equilibrium” and “balanced” to explain your model or result. Sentence starters are optional.

equilibrium: zero net force and zero acceleration

balanced: adding to zero net force in this force context

tension: the pulling force through a stretched rope or cable

01 Warm-up: bring back an earlier idea

A 3 kg box slides left at 5 m/s on a rough horizontal floor after release.
Draw forces and explain what follows about vertical and horizontal acceleration.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

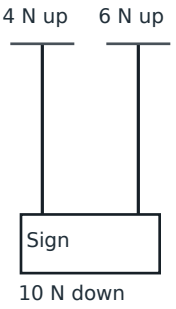
02 Worked example: explain the model

Reference: $\Sigma F_x = 0$; $\Sigma F_y = 0 \Rightarrow a = 0$.
Equilibrium allows rest or constant velocity. Resolve missing forces along each axis. Use g = 9.8 N/kg.

Worked reasoning: A sign has weight 10 N down and two vertical upward pulls of 4 and 6 N. The vector sum is zero, so equilibrium is possible.

Your turn: Draw and label all three forces.
Identify the equation that checks vertical balance.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

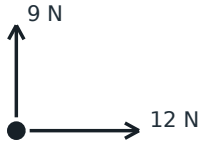
Sentence starters are optional. You may plan in any language.

03 Two component balance

A stationary object has 12 N right and 9 N up. A third force balances both.
Find its components, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Third force not shown

Marzano check - circle: 1 2 3

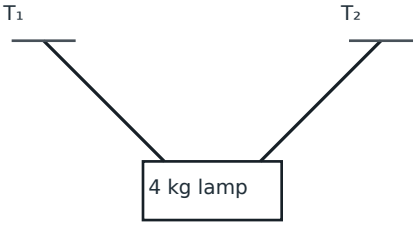
04 Two-rope hanger

A stationary 4 kg lamp hangs from two identical ropes, each 45° above horizontal.
Use g = 9.8 N/kg. Resolve the tensions into components and find each tension.
Explain why each tension is larger than half the weight.

Word help: resolve = separate a vector into components

Sentence starter: Both … . They differ in … because … .

45° above horizontal



Marzano check - circle: 1 2 3

05 Change angle

A 10 kg sign hangs from two identical ropes.
Compare ropes 30° above horizontal with ropes 60° above horizontal.
Use g = 9.8 N/kg.
Sketch both arrangements, calculate each tension, and explain which puts more tension in the ropes.

Sentence starter: Both … . They differ in … because … .

First arrangement: 30°



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A=(7, -5) m and B=(-5, 9) m.
Find R, magnitude and direction.
Choose C so A+B+C=0 and sketch the route.
Draw the component vectors and resultant; label lengths.

Word help: component = the part of a vector along one chosen axis

Sentence starter: I found … [unit]. This value means … in this situation.

North / +y (m)



East / +x (m)

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 3 kg lamp hangs at rest from a vertical rope.
Use g = 9.8 N/kg.
Find tension. Then redraw the forces if the lamp accelerates upward. Write the equation showing why tension must change.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Add and label force arrows.

Choose the lamp as the object.

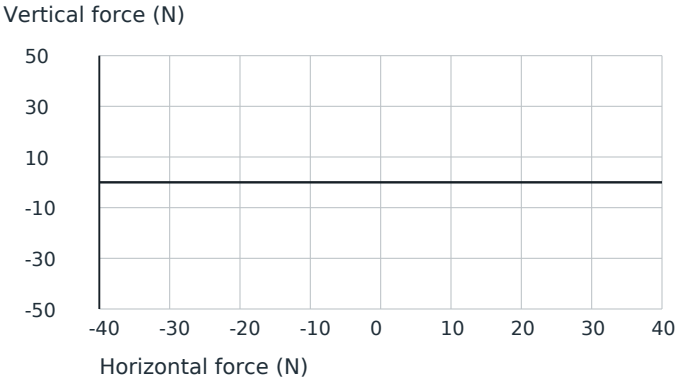
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A stationary junction has a 30 N horizontal force right, a 40 N vertical force down, and one unknown force.
Find the unknown vector, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3